

§1 Jacobian $dx dy = r dr d\theta$

Green $\oint_L \frac{-y dx + x dy}{x^2 + y^2} = 2\pi$

$E_n^{(1)} = \langle n^{(0)} | H' | n^{(0)} \rangle$

$E_n^{(1)} = \sum_{m \neq n} \frac{\langle m^{(0)} | H' | n^{(0)} \rangle^2}{E_n^{(0)} - E_m^{(0)}}$

$a^\dagger = \sqrt{\frac{m\omega}{2\hbar}} (x - \frac{i p}{m\omega})$

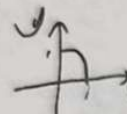
$a = \sqrt{\frac{m\omega}{2\hbar}} (\bar{x} + \frac{i p}{m\omega})$

$\bar{x} = \sqrt{\frac{\hbar}{2m\omega}} (a^\dagger + a)$

$p = i\sqrt{\frac{m\hbar\omega}{2}} (a^\dagger - a)$

$\bar{x} |n\rangle = \sqrt{\frac{\hbar}{2m\omega}} (\sqrt{n} |n-1\rangle + \sqrt{n+1} |n+1\rangle)$

$\bar{x} |n\rangle = \sqrt{\frac{\hbar}{2m\omega}} (\sqrt{n} |n-1\rangle + \sqrt{n+1} |n+1\rangle)$

§2 M1 

$I = \iint_D \frac{x^2 + \sin(xy) + x^2 y}{x^2 + y^2} dx dy$

$= \iint_D \frac{x^2}{x^2 + y^2} dx dy + \iint_D \frac{\sin(xy)}{x^2 + y^2} dx dy + \iint_D \frac{x^2 y}{x^2 + y^2} dx dy$

$= \iint_D \frac{x^2 + y^2}{x^2 + y^2} dx dy + \iint_D \frac{\sin(xy)}{x^2 + y^2} dx dy$

$= \iint_D 1 dx dy + \iint_D \frac{\sin(xy)}{x^2 + y^2} dx dy$

$= \int_0^{\pi/2} \int_0^1 r dr d\theta + \iint_D \frac{\sin(xy)}{x^2 + y^2} dx dy$
 $= (\frac{1}{2} + \frac{1}{2}) \cdot \frac{1}{2} = \frac{1}{2}$

Q1 (i) $E_n^{(1)} = \langle n^{(0)} | H' | n^{(0)} \rangle = \langle n^{(0)} | \lambda(x^2 y + x y^2) | n^{(0)} \rangle = \lambda \langle 000 | \lambda(x^2 y + x y^2) | 000 \rangle$

宇称反演 $x \rightarrow -x, y \rightarrow y$
 $H'(x, y) = -H'(-x, y)$ 奇函数
 $= \lambda \langle 00 | x^2 | 00 \rangle \langle 0 | y | 0 \rangle + \lambda \langle 00 | x | 00 \rangle \langle 0 | y^2 | 0 \rangle$
 $= \lambda \cdot 0 \cdot 0 = 0$

(II). $E_n^{(1)} = \sum_{m \neq n} \frac{|\langle m^{(0)} | H' | n^{(0)} \rangle|^2}{E_n^{(0)} - E_m^{(0)}} \quad \frac{1}{2} C \equiv \sqrt{\frac{\hbar}{2m\omega}} \quad x = C(a + a^\dagger), y = C(a_y + a_y^\dagger)$

$x y |000\rangle = C^2 (|10\rangle + \sqrt{2} |20\rangle) |1\rangle_y |0\rangle_z = C^2 (|010\rangle + \sqrt{2} |210\rangle)$

$y^2 x |000\rangle = C^3 \sqrt{2} |100\rangle + \sqrt{2} |300\rangle$

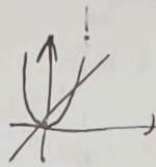
$|100\rangle \text{ 与 } |010\rangle \text{ 正交}$
 $\Delta E = \hbar\omega \sum |\langle m^{(0)} | H' | n^{(0)} \rangle|^2 = 2\lambda^2 C^6 \quad E_n^{(1)} = -\frac{\langle 00 | C^6}{3\hbar\omega} = -\frac{5\sqrt{2}\hbar^4}{16m^2\omega^4}$

$|200\rangle \text{ 与 } |120\rangle \text{ 正交}$
 $\Delta E = -3\hbar\omega \quad \sum |\langle m^{(0)} | H' | n^{(0)} \rangle|^2 = 4\lambda^2 C^6$

(III). $E_n^{(1)} < 0$ 下移. 与原有能量态向下移动有联系

§3

(15) $I_1 = \oint_C (2xy - y^2)dx + (x^2 + 2xy)dy$



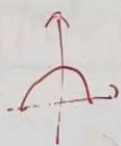
$$= \int_0^1 \int_{x^2}^1 (2x + 2y - y^2) dy dx$$

$0 \leq x \leq 1$

$y = x^2$

$$= \int_0^1 dx \int_{x^2}^1 (2x + 2y - y^2) dy = \frac{4}{15}$$

(17)



路径: 线 $L_1: y=0$, x 从 -1 到 1 .

$$L_1 = \int_{-1}^1 (e^x \sin y - y)dx + (e^x \cos y - 1)dy$$

闭曲线 $C = L_1 + L_2$

$$\frac{\partial Q}{\partial x} = \cos y e^x \quad \frac{\partial P}{\partial y} = e^x \cos y = 1$$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1$$

$$\oint_C P dx + Q dy = \iint_D 1 dx dy = \frac{\pi}{2}$$

$$\int_{L_1} = \int_{-1}^1 0 dy = 0 \quad \int_{L_2} = \oint_C - \int_{L_1} = \frac{1}{2} \pi$$

(18)

$$I_3 = \oint_C \frac{xy - y^2}{x^2 + y^2} (x+y) dx + (x-y) dy$$

C : 椭圆 $\frac{x^2}{4} + \frac{y^2}{9} = 1$. 逆时针.

$$I_3 = I_{3A} + I_{3B}$$

$$I_{3A} = \oint_C \frac{xy - y^2}{x^2 + y^2} dx$$

$$I_{3B} = \oint_C \frac{(x+y) dy}{x^2 + y^2} = \oint_C \frac{(x+y) dy}{x^2 + y^2}$$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1 - 1 = 0 \Rightarrow I_{3B} = 0$$

对 I_{3A} 取奇点. 椭圆绕原点. $x^2 + y^2 = \epsilon$ (足够小)

$$I_{3A} = 0 + \oint_C \frac{xy - y^2}{x^2 + y^2} = 2\pi$$

$$I = I_{3A} + I_{3B} = 2\pi$$